



# Airbnb Data Analysis - Excel

# GREETINGS

Hello, It's Pernel.

*I was very delighted to work on this project as it aligned perfectly with my passion for finding innovative solutions to business challenges. This opportunity allowed me to design and engage in an innovative sector, explore business constraints, leverage my data skills, and satisfy my curiosity by applying a data-driven approach at every step.*

*I hope that each reader will enjoy reading this report as much as I enjoyed creating it.*

## Stay Hysinghts !

NB: It is important to understand that all data gathered in this context are influenced by specific circumstances so that it may not accurately represent reality.

# Summary .

- 1 About the project
- 2 Project Workflow
- 3 Recommendations

# ABOUT THE PROJECT

## Business Description

Airbnb is a global online marketplace that connects hosts offering accommodations with guests seeking short-term lodging experiences. Founded in 2008, Airbnb has revolutionized the travel and hospitality industry by allowing individuals to rent out their homes, apartments, or unique spaces. The platform facilitates transactions, provides secure payment options, and enables both hosts and guests to rate and review each other.

Airbnb operates as a two-sided marketplace, with key stakeholders including hosts (property owners), guests (travelers), and the platform itself. The business model is primarily commission-based, earning fees from both hosts and guests for each booking.



# ABOUT THE PROJECT

## Project Overview

The dataset includes details about hosts, locations, pricing, and key metrics, aiming to uncover trends in guest preferences, pricing strategies, and market dynamics across various European cities. The analysis helps stakeholders make informed decisions about the accommodation, expenses, and preferred room types in specific areas. It also guides travellers them in finding ideal locations based on specific criteria like price or hosts' reputation, making trip planning easier for those unfamiliar with the destinations.

## Questions

1. Which factors have the greatest impact on the pricing of listings in various European cities ?
  2. How do guest preferences differ from one city to another ?
  3. Which areas within the cities experience the highest demand, and what features contribute to their appeal for guests ?
  4. What market trends and seasonal variations are observed in occupancy rates and pricing across different cities ?
  5. What insights can be drawn about future booking trends and potential revenue expansion for hosts ?
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# PROJECT WORKFLOW

# ABOUT DATASET

We created our dataset by gathering information from various Airbnb listings in cities, with a focus on certain cities that have exhibited significant activity from the company. We selected features that aligned with our analysis goals. Initially, the raw data was organized by region and split into time periods (weekdays and weekends) before being combined into a single comprehensive dataset for better analysis and understanding.

The dataset includes about 51,707 listings with 22 columns, featuring both categorical and numerical data. The cities represented are Amsterdam, Athens, Barcelona, Berlin, Budapest, Lisbon, London, Paris, Rome, and Vienna. The Booking prices range from \$34.78 to \$18,545.45 per night (Note: The pricing format on Airbnb is \$/night, which has been considered in this analysis).

There are three types of room categories : Shared Room, Private Room, and Entire home/Apt. Notably, Shared Rooms are the least popular, despite being generally cheaper.

**Please Take a look at the metadata file to explore all the features**



# CLEANING & PREPARATION

## SUMMARY

- *Missing Data Handling*: No value found
  - *Handling Duplicates* : No Duplicates streaming found
  - *Standardization*: All boolean Data has been turned into numerical values to better analysis process
  - *Features*: room\_shared and room\_private have been delete because their information is already present on room\_type
  - *Outlier Treatment*: Some data observed in booking amount field are too high but at this point of the analysis, we can tell exactly that if there are outliers or not.
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# CLEANING & PREPARATION

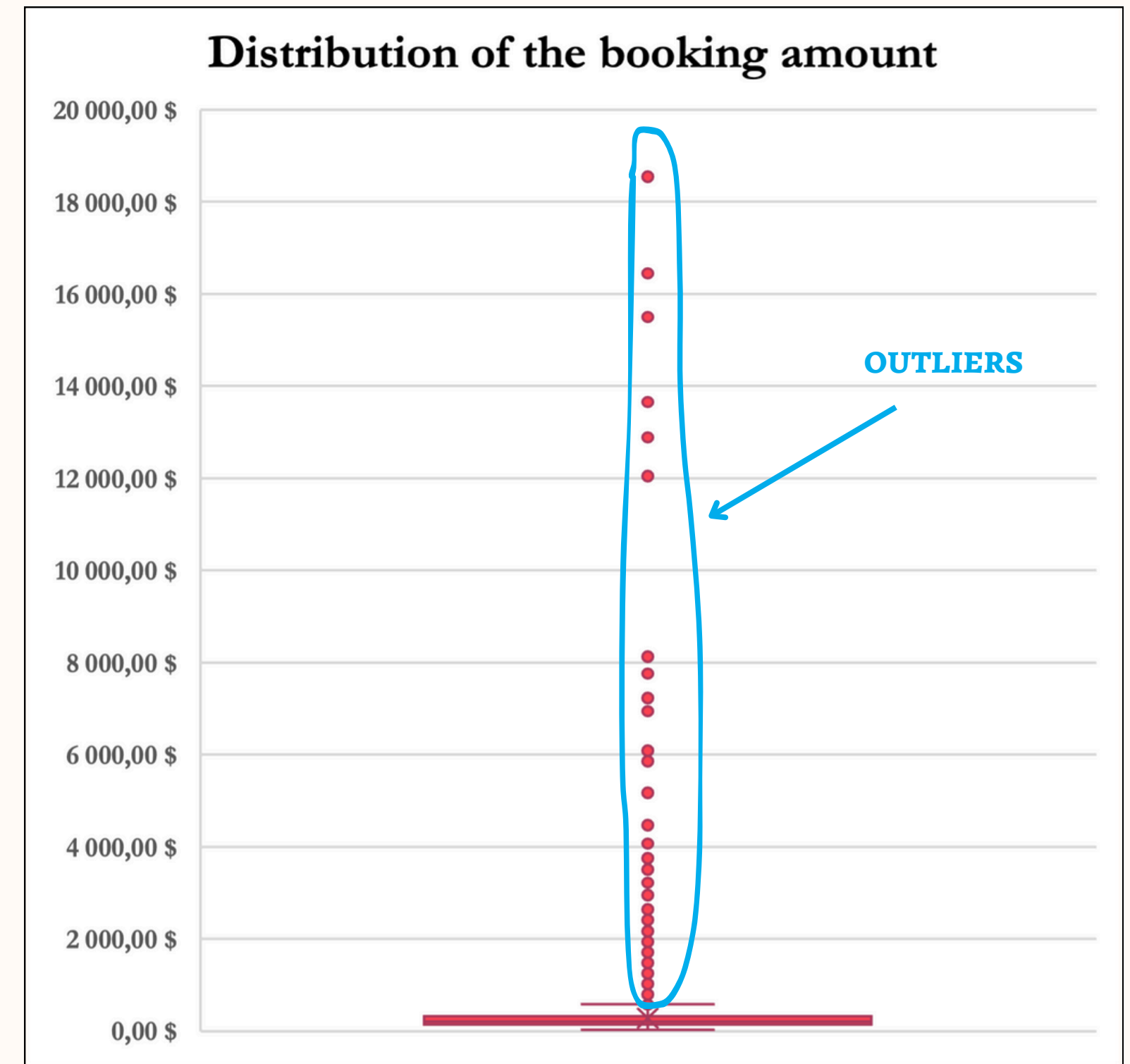
## HANDLING OUTLIERS

The boxplot of booking amount suggests that the distribution is *extremely right-skewed*.

|        |        |
|--------|--------|
| Mean   | 279,88 |
| Median | 211,34 |
| Std    | 327,95 |

The descriptive stats observed suggests that the price mostly concentrate around 211\$ (as median represents 50% of the data).

The surrounded data points might be considered as outliers. The values like 18,000\$ are *extremely abnormal (outlier)* for a one night renting price.



# CLEANING & PREPARATION

## AFTER HANDLING OUTLIER ANALYSIS

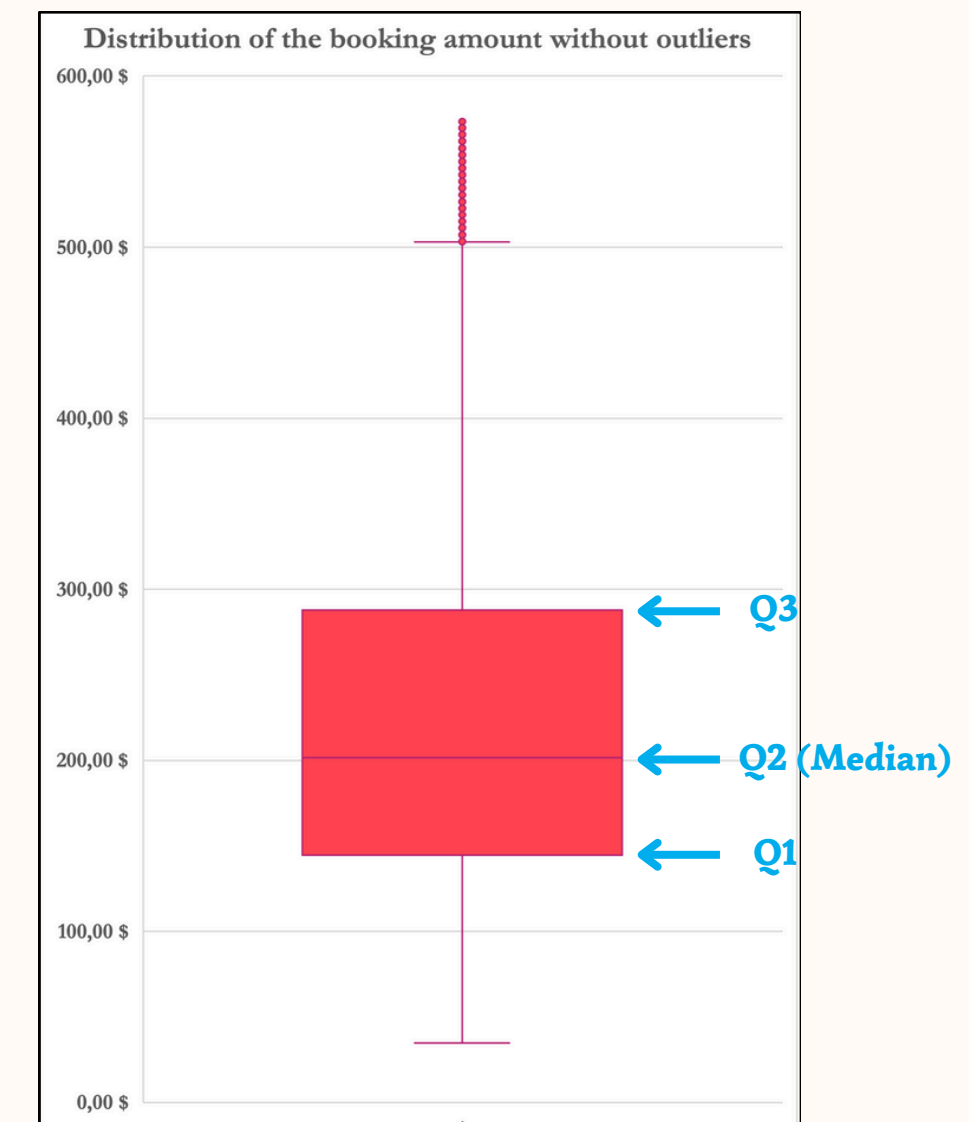
After conducting a visual analysis using Boxplot Interpretation, we apply the Interquartile Range (IQR) method to identify the outlier values, representing around 7% of the dataset. Given the uncertainty regarding the legitimacy of the high prices, we are considering the exclusion of these outlier data points.

The data is more readable, considered as slightly right-skewed (median is closer to Q1). This indicate that most of the airbnb are approximatively booked at **200\$/night**. The IQR (range Q1 to Q3) suggests that most airbnb are clustered around the **144 \$ - 288 \$** range.

The presence of higher amount may indicates exceptional performances or may due to other conditions like **room\_type** or **person\_capacity**.

IQR Stats

|              |         |
|--------------|---------|
| IQR          | 170,94  |
| 1st quartile | 148,75  |
| 3rd quartile | 319,69  |
| Lower Bound  | -107,66 |
| Upper Bound  | 576,11  |

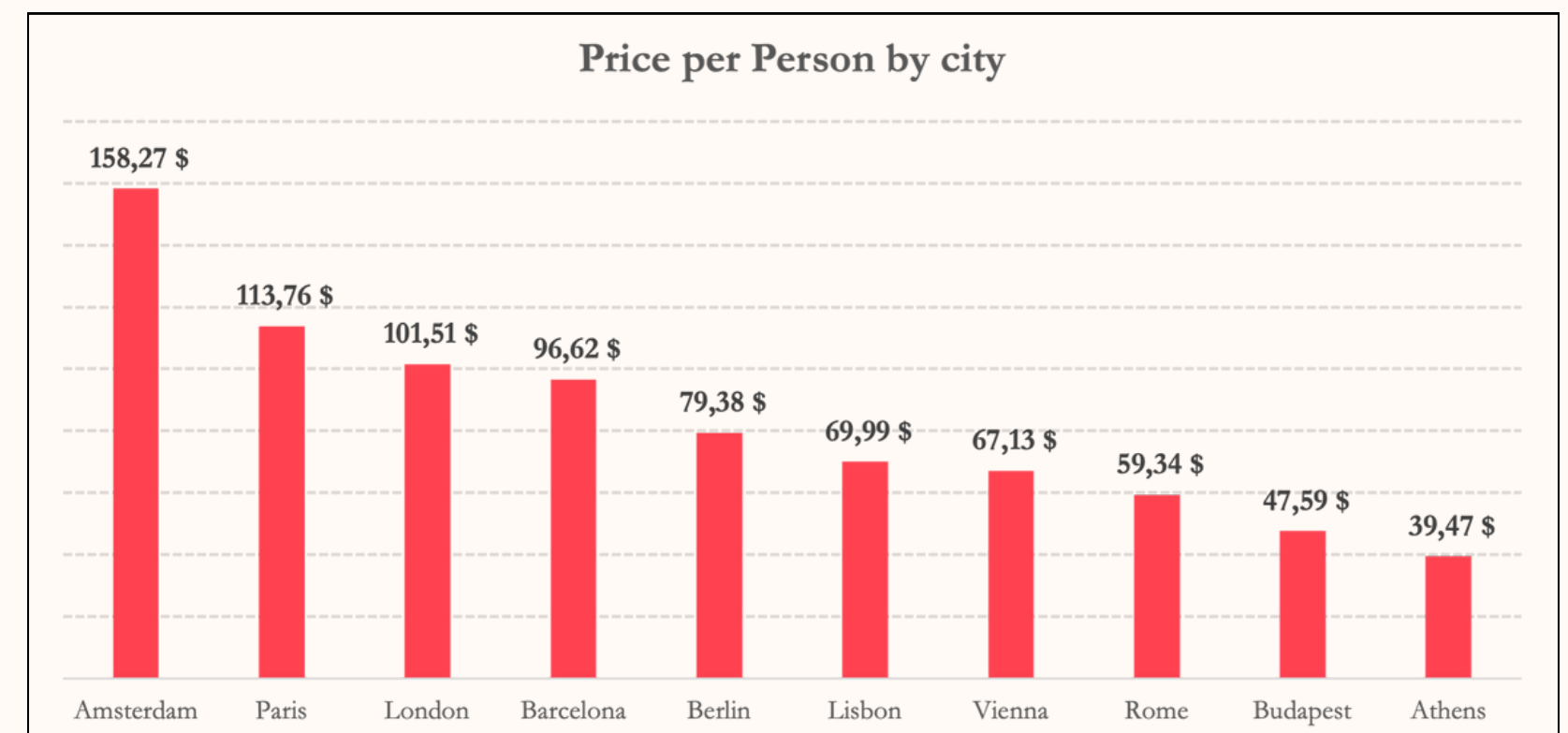
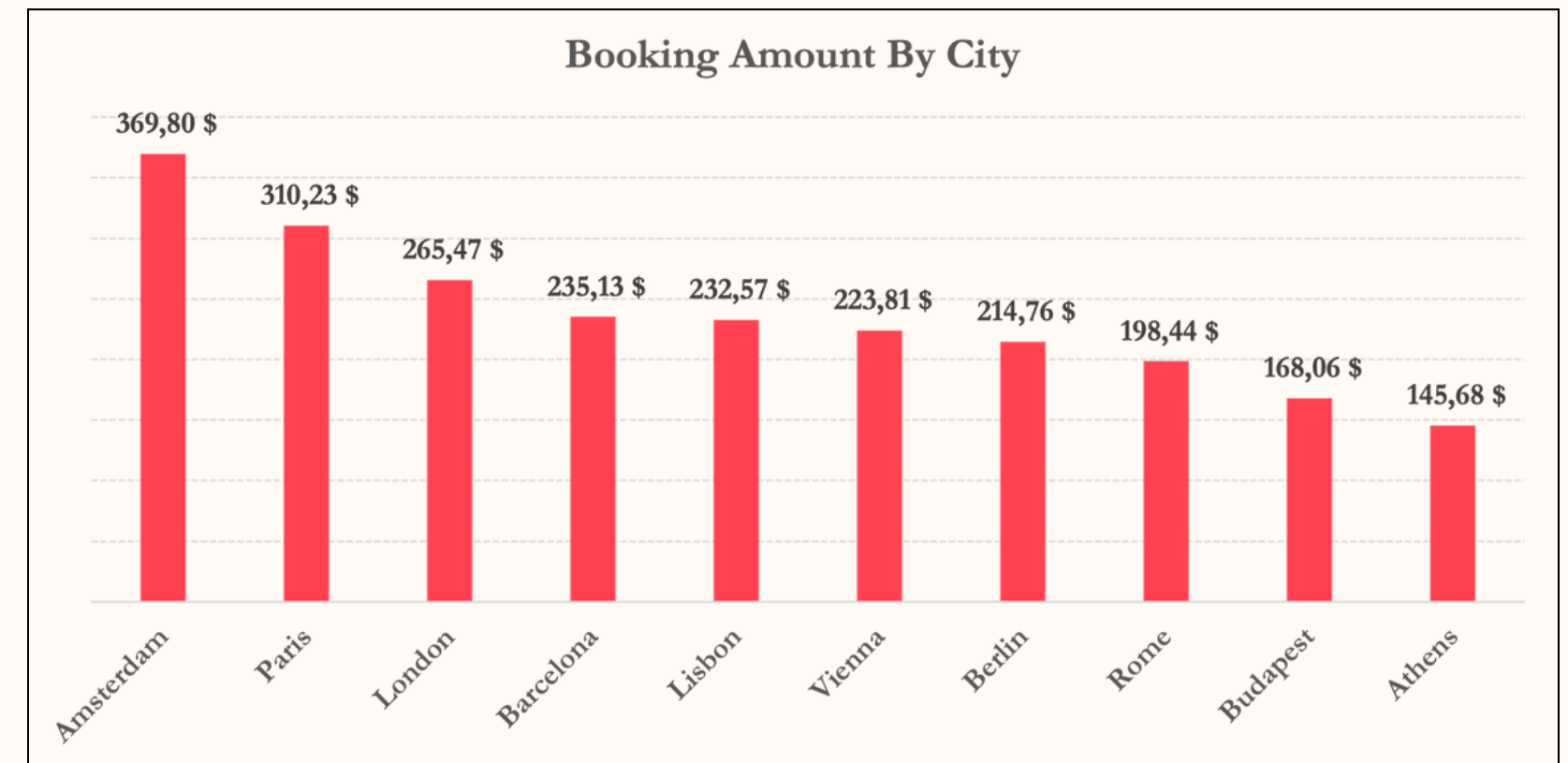


# ANALYSIS & VISUALIZATION

## MARKET POSITIONING

The data suggests that guests heading to Amsterdam or Paris should prepare for higher accommodation costs, while those visiting Athens or Budapest might find more budget-friendly options.

Further analysis of factors such as property features, amenities, and seasonal demand could provide additional insights into these differences.



# ANALYSIS & VISUALIZATION

## SATISFACTION IMPACT ON PRICING

This heatmap displays the average guest satisfaction ratings across various European cities.

The color gradient from red to green identifies cities with lower to higher satisfaction scores. Despite lower booking prices, cities like Athens and Budapest still excel in guest satisfaction potentially offering better value for money.

On the other hand, higher-priced cities like London and Paris, while still performing well, may need to address specific areas to enhance guest experiences further.

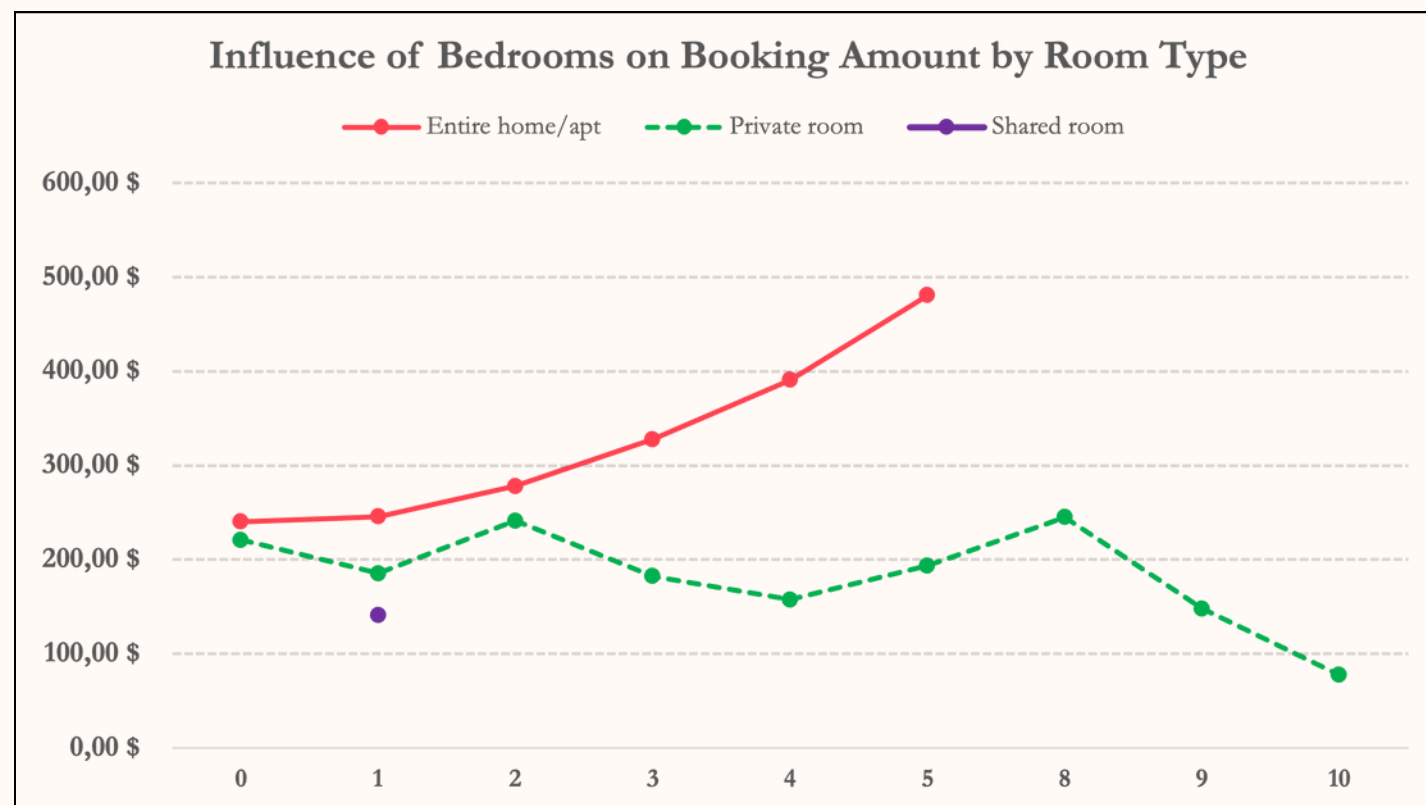
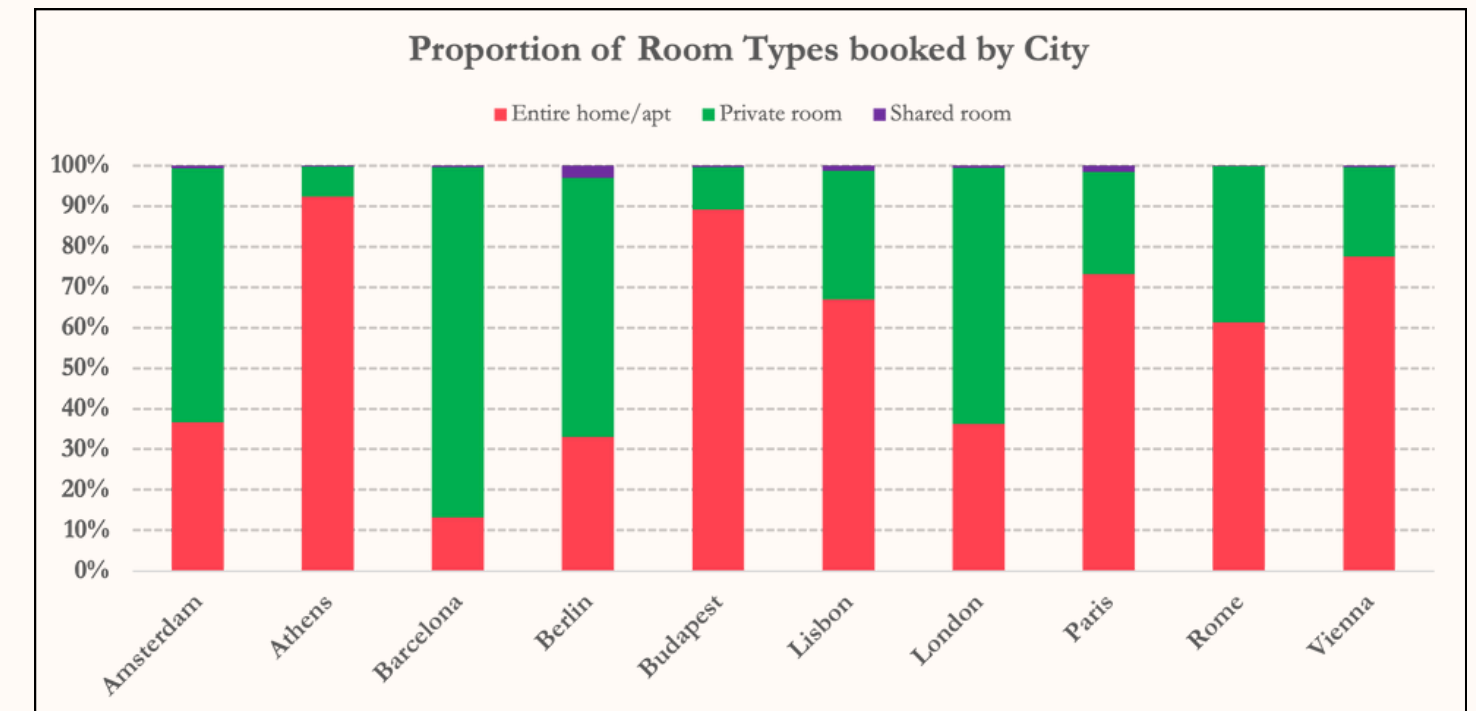
| Cities    | Avg_guest_satisf |
|-----------|------------------|
| London    | 90,60            |
| Lisbon    | 91,04            |
| Barcelona | 91,24            |
| Paris     | 91,84            |
| Rome      | 93,10            |
| Vienna    | 93,73            |
| Amsterdam | 94,1             |
| Berlin    | 94,33            |
| Budapest  | 94,60            |
| Athens    | 94,99            |

# ANALYSIS & VISUALIZATION

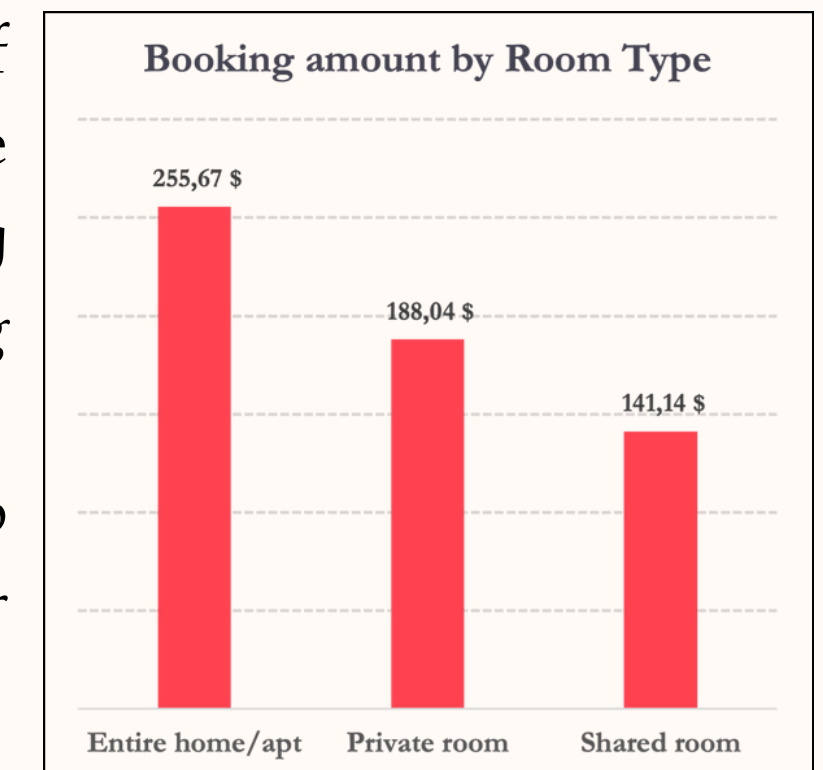
## PRICE AND ROOM TYPE IMPACT ON PRICING

**Entire Home/Apt** is the preferred choice for guests and commands the highest booking amounts. **Private room** is also a popular option but offer a more affordable option compared to entire home.

**Shared Room** despite being the cheapest pricing option is rarely booked. This may suggests that intimacy is really important for guests.



Based on the analysis of the influence of bedrooms on Pricing, it appears that the number of bedrooms has a **strong positive influence** on the booking amount for **entire home/apt** properties. This suggests that guests are willing to pay more for properties with a larger number of bedrooms.





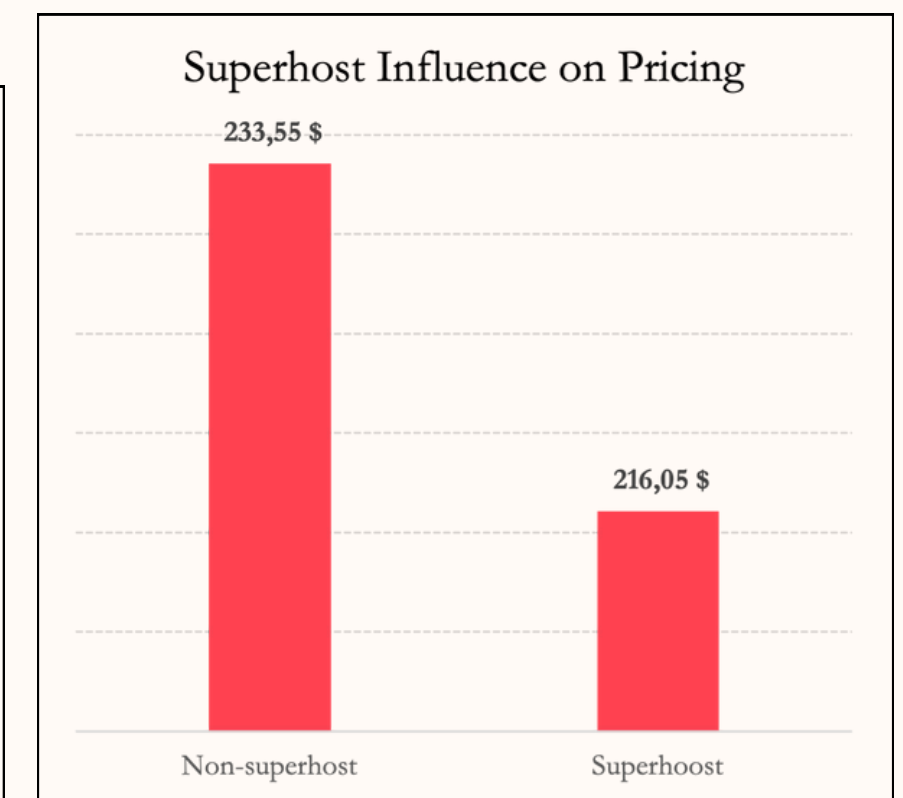
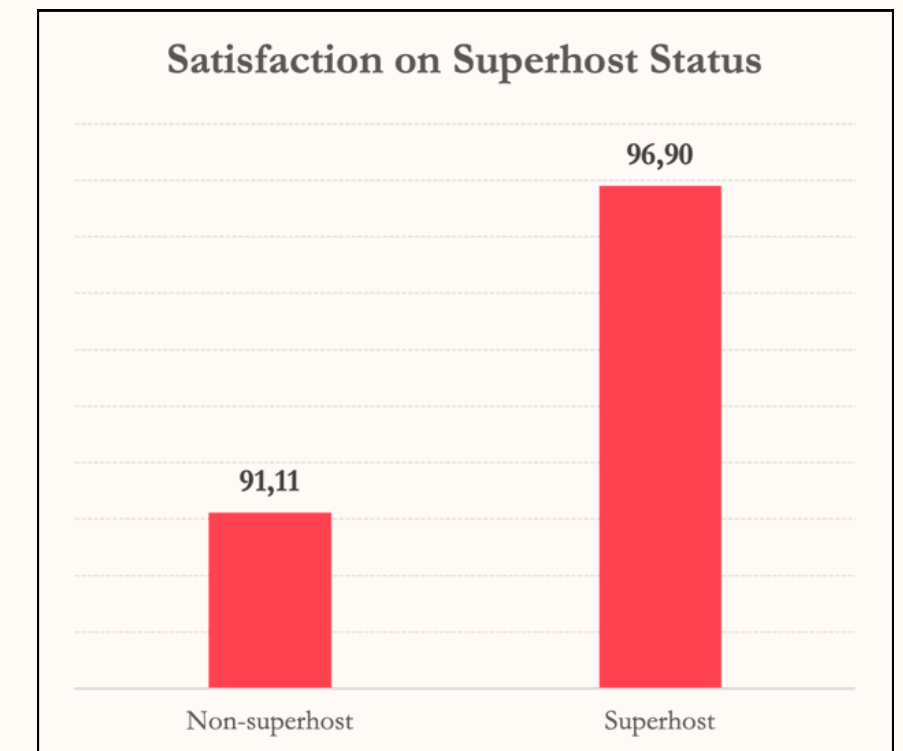
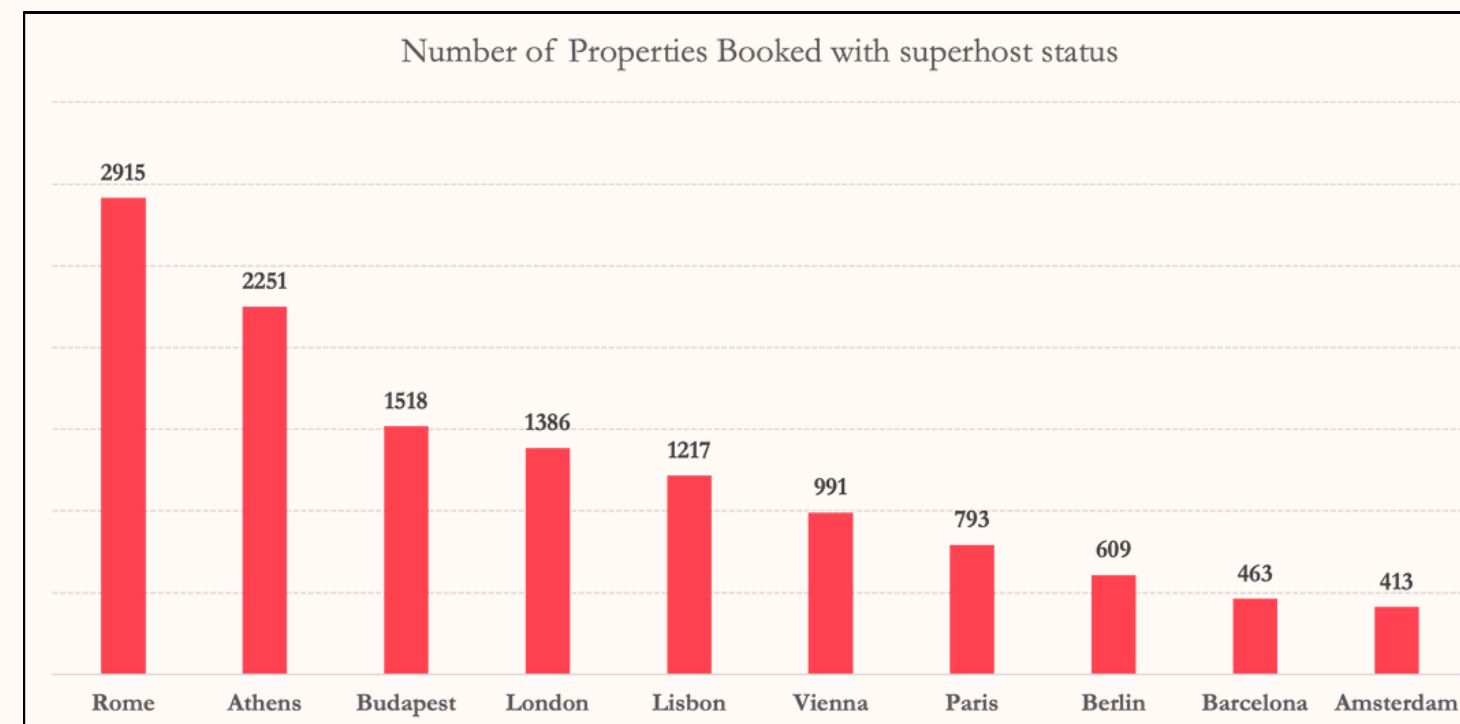
# ANALYSIS & VISUALIZATION

## SUPERHOST IMPACT ON PRICING

Contrary to the assumption that **Superhost** would set higher prices owing to their esteemed reputation, the data indicates that **Non-Superhost**, on average, charge more.

This may suggest that **Superhost** focus on alternative priorities, such as achieving higher occupancy rates or enhancing customer satisfaction, rather than solely maximizing their nightly rates. Moreover, **Rome, Athens and Budapest**, knows as the most properties with Superhost Status, generally command budget friendly properties.

Evaluating the influence of Superhost status reinforces the notion that guests is greater satisfied when their Airbnb host is a Superhost.

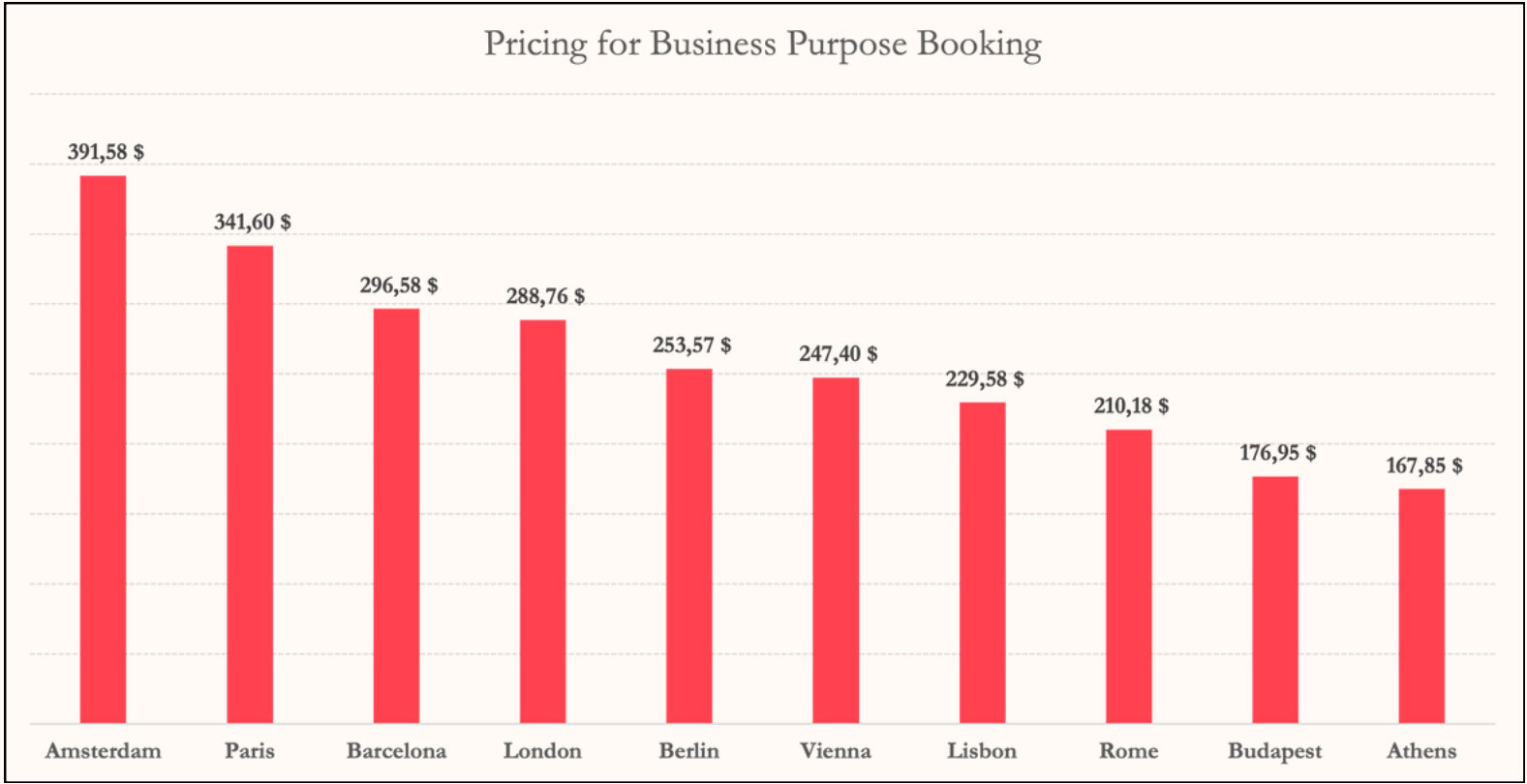
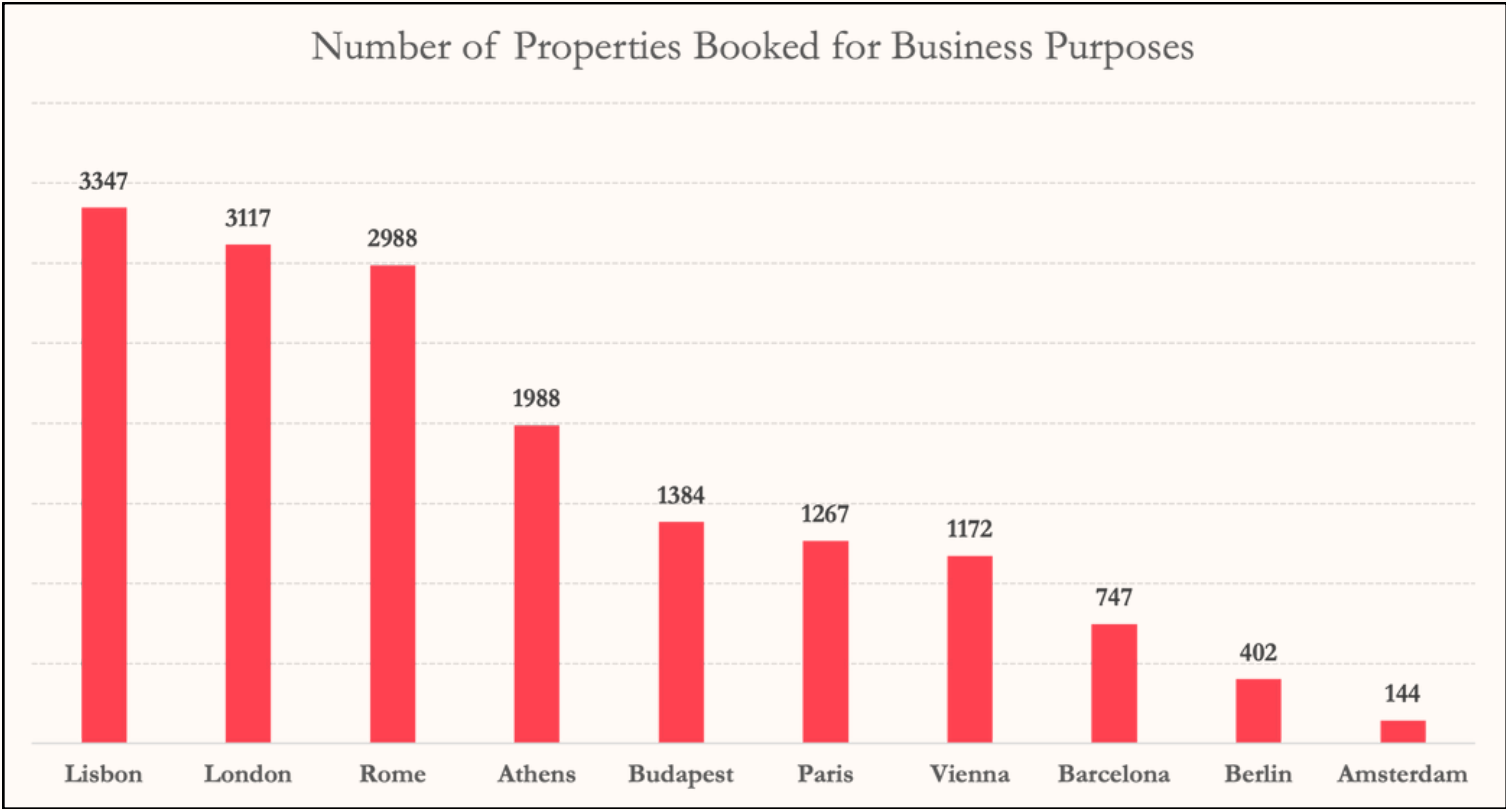
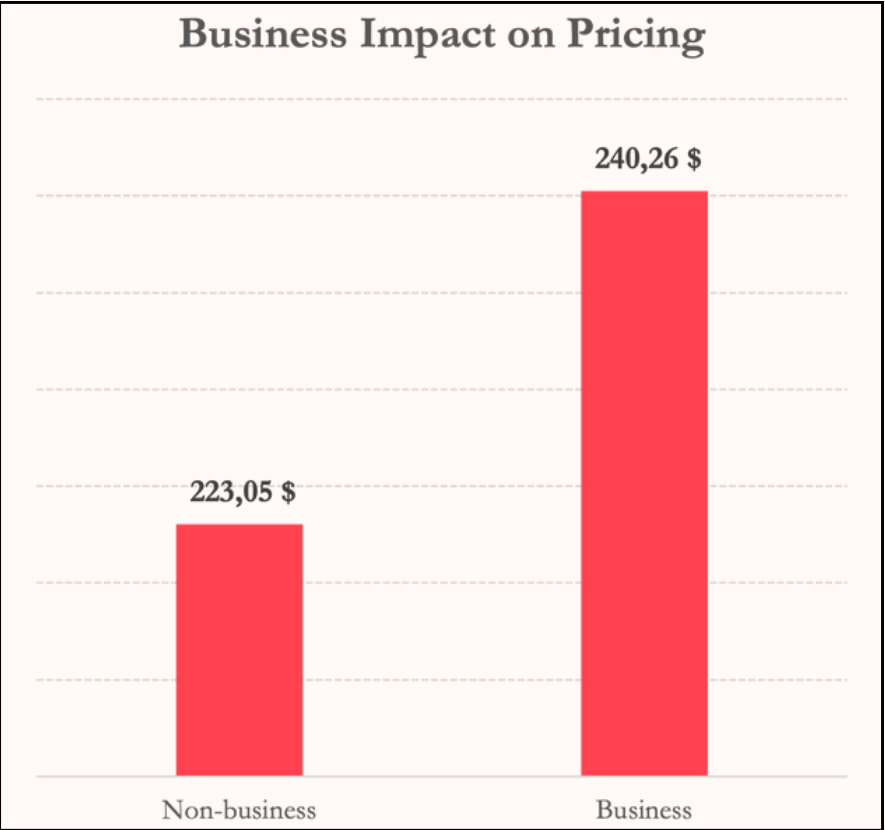


# ANALYSIS & VISUALIZATION

## BUSINESS IMPACT ON PRICING

In general, properties designated for business use are usually a bit more costly compared to other properties. The cities of Lisbon, London, and Rome have the highest rates of business-related bookings. However, these cities often present reasonable rates for guests.

Amsterdam, Paris and Barcelona command higher prices of business-related bookings, that correlates with the lower rates of business-related bookings.

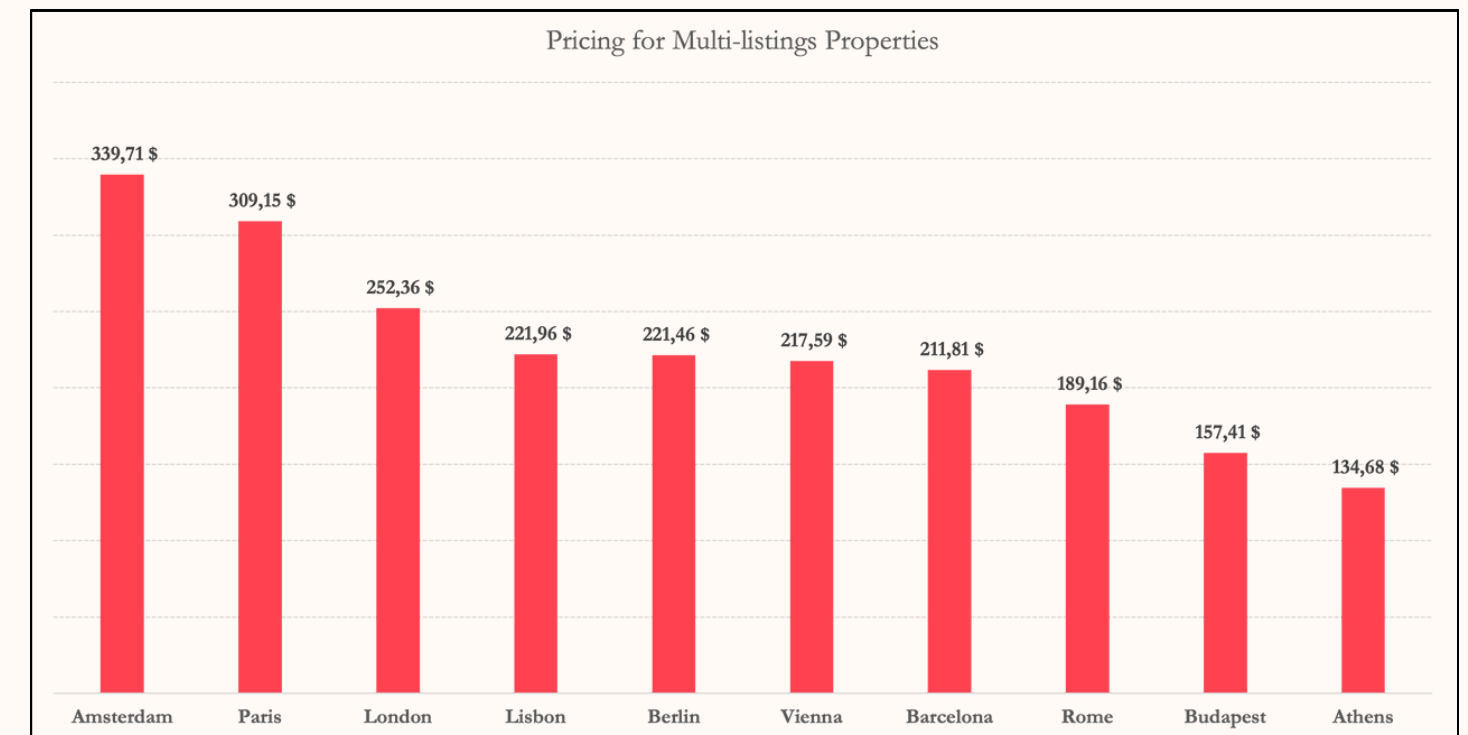
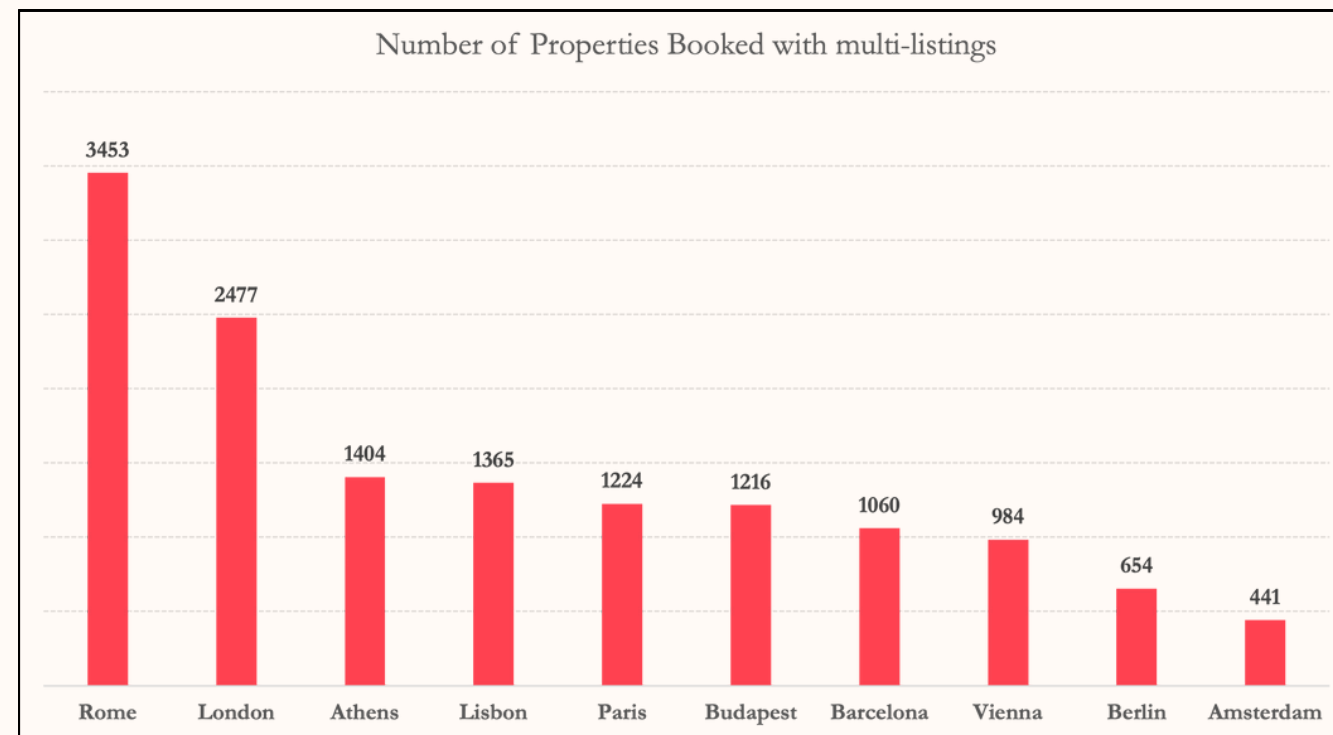
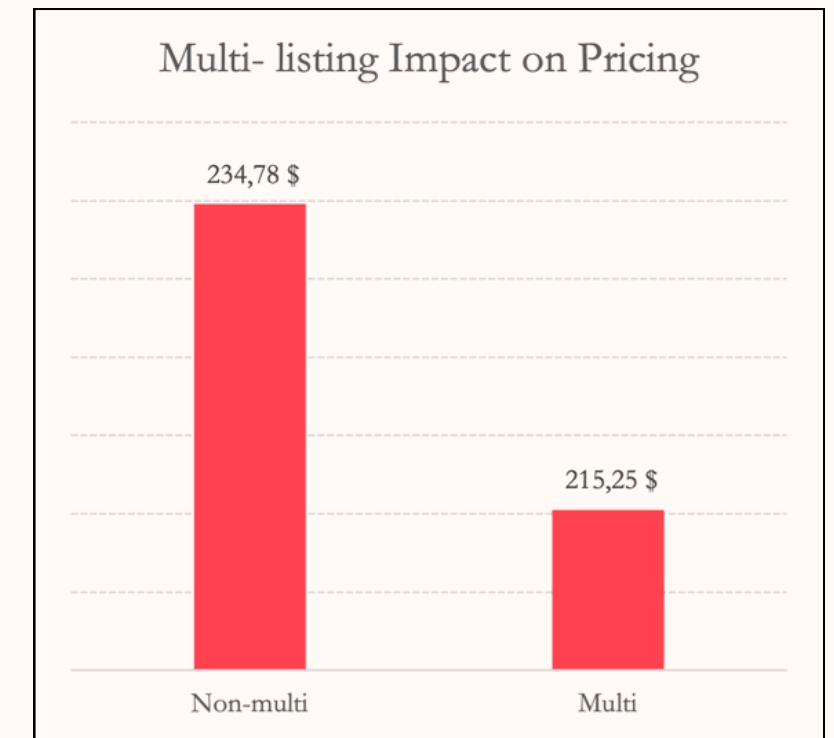


# ANALYSIS & VISUALIZATION

## MULTI-LISTINGS IMPACT ON PRICING

Contrary to the assumption that **Multi-listings Properties** would set higher prices, the data indicates that **Non-multi listings** properties charge more.

By inspecting the number of properties booked with multi-listings, we find that Rome and London attract guests with multi- listing needs contrary to Amsterdam that attract less. But Amsterdam still command the highest price.





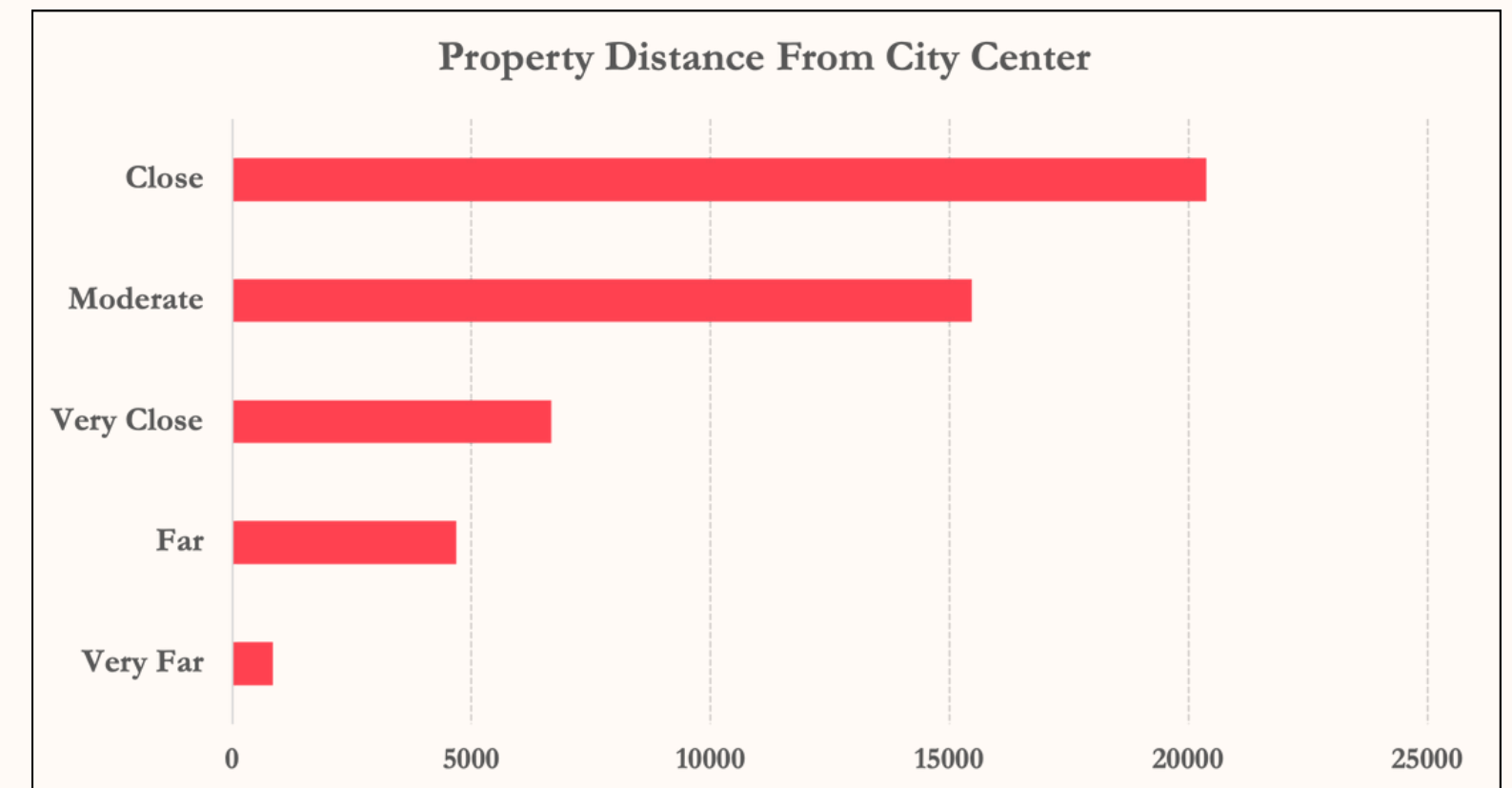
# ANALYSIS & VISUALIZATION

## LOCATION AND ACCESS IMPACT

In this section, we introduce four new features designed to enhance the categorization of distance-related insights: **distance\_name**, **distance\_metro\_name**, **distance\_attr\_name**, and **distance\_rest\_name**. These features classify properties into six categories: “**Very Close**”, “**Close**”, “**Moderate**”, “**Far**”, “**Very Far**” and “**Extremely Far**” according to their respective distances.

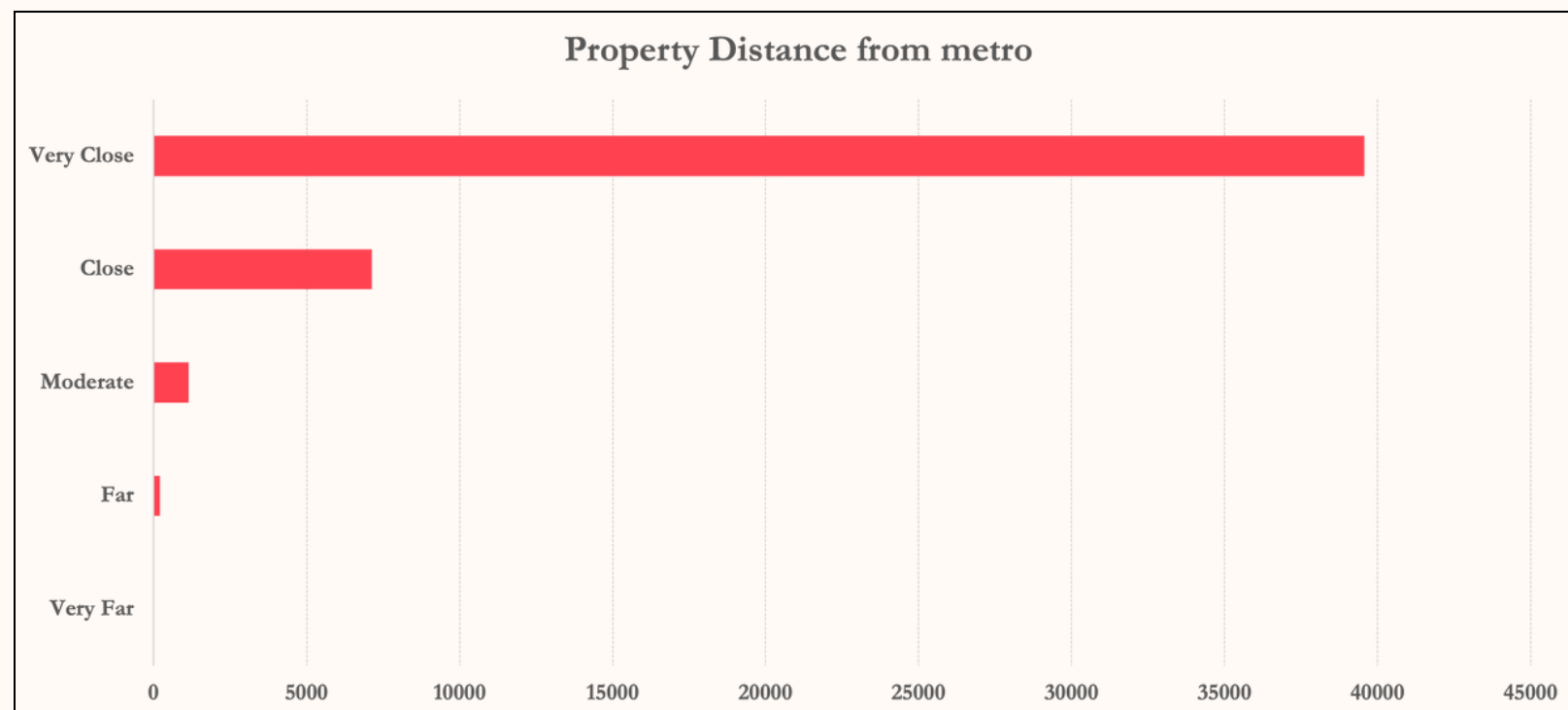
### Property Distance From City Center

The majority of properties are located at a close distance (between 1 and 3 miles) from the city, with fewer properties located far or very far.



### Property Distance From Metro

The majority of properties are located at a very close (up to 1 mile) distance from the metro, with fewer properties located far.

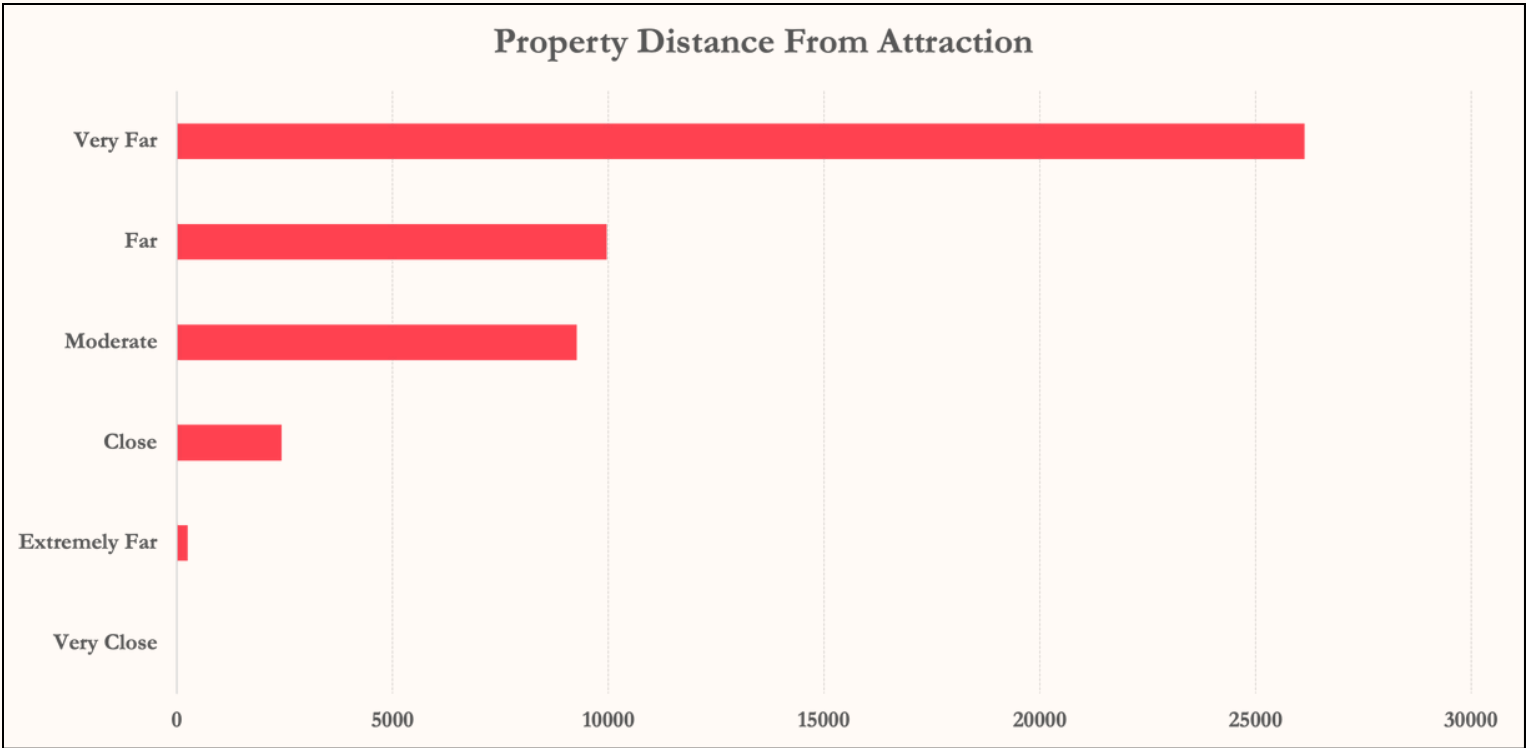
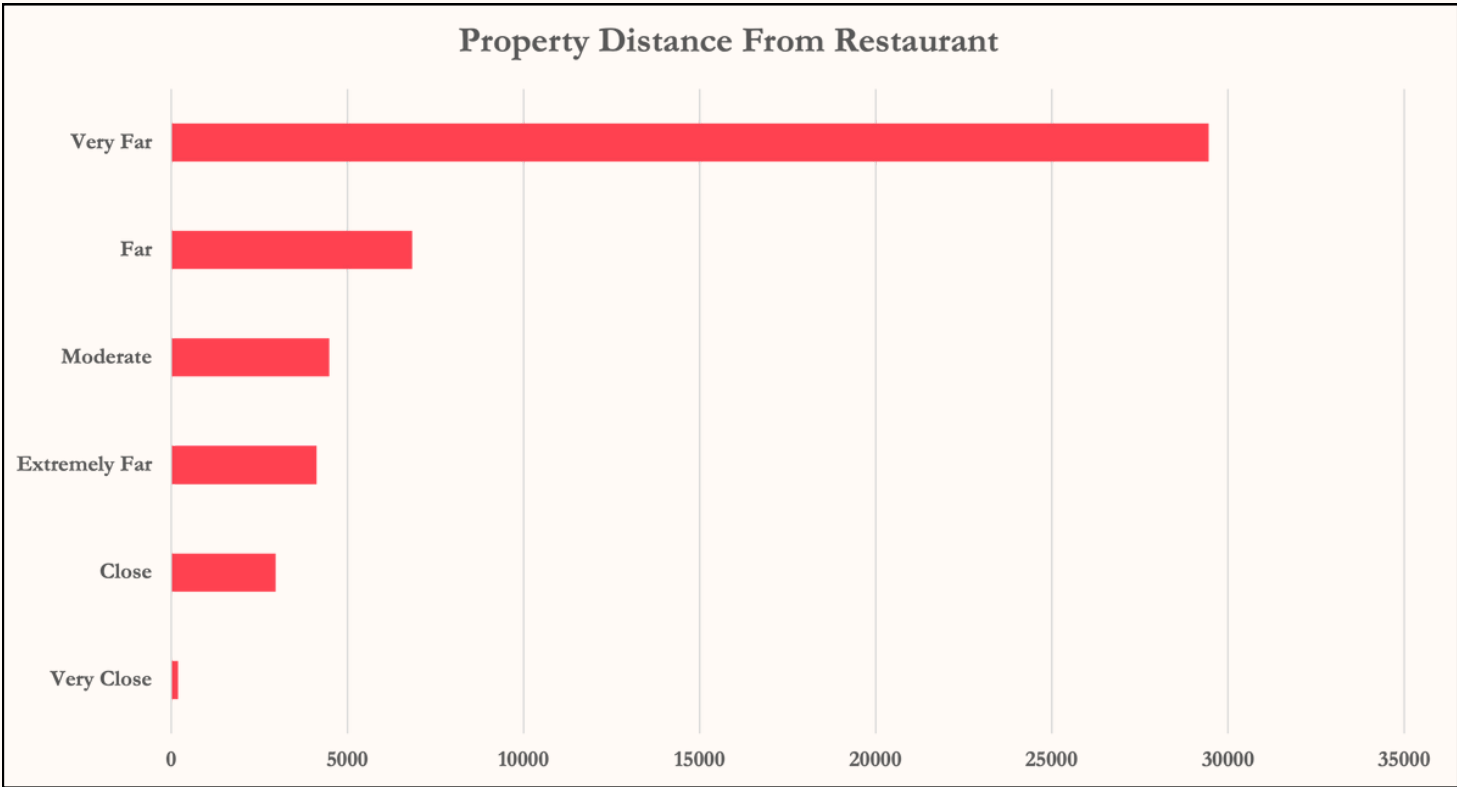


# ANALYSIS & VISUALIZATION

## LOCATION AND ACCESS ANALYSIS

### Property Distance From Restaurant

The majority of properties are located at a very far distance (more that 50 miles) from attractions, with fewer properties located very close or close.



### Property Distance From Attraction

The majority of properties are located at a very far distance (more that 50 miles) from attractions, with other properties located relatively closed.

Interestingly, even at greater distances, some listings still achieve high booking prices, suggesting that proximity to distance factor is not the only factor influencing pricing.

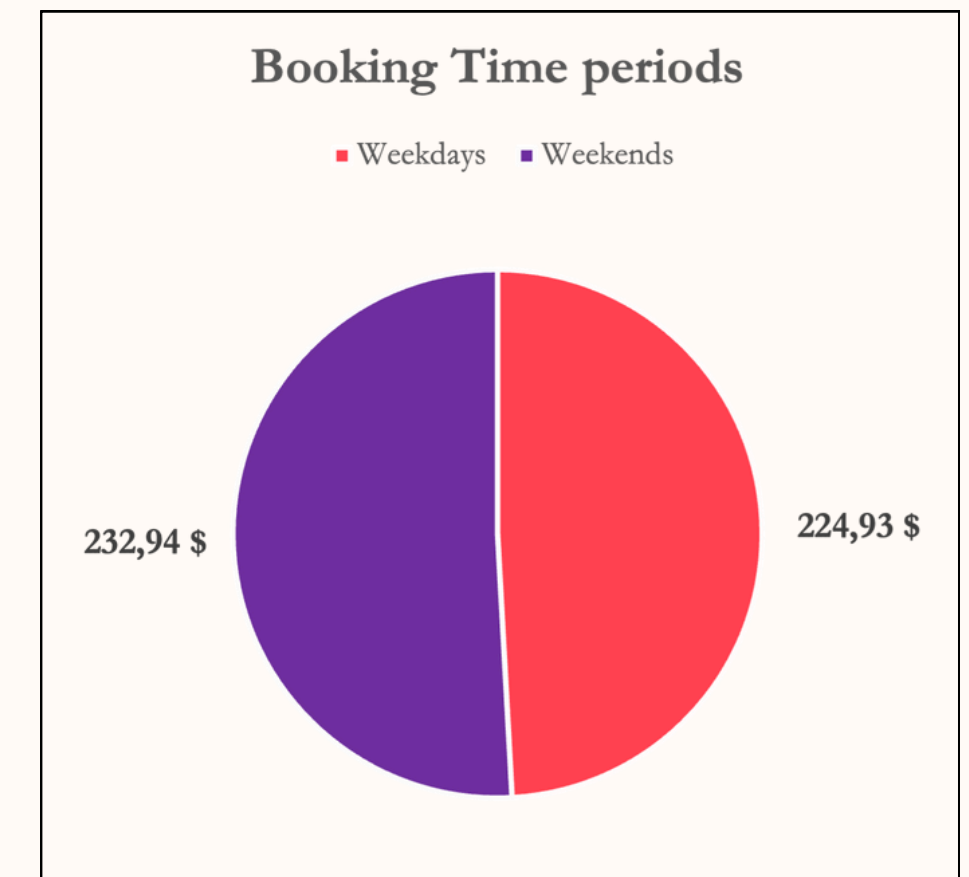
# ANALYSIS & VISUALIZATION

## TIME PERIOD ANALYSIS

This pie chart illustrates the distribution of pricing differences between weekdays and weekends.

While weekends traditionally are seen as more valuable booking days, the data suggests only a slight advantage in pricing, pointing towards a market where weekday and weekend demand may be closer than expected.

Actually, we don't have this data but we rather thinking that a deep dive in specific time periods or season analysis could reveal patterns to support more dynamic pricing variation of weekdays and weekends.



**Please Take a look at the dashboard section on Excel file to explore the report**

# RECOMMENDATIONS

# AIRBNB ACTIVITY MARKET

Which factors have the greatest impact on the pricing of listings in various European cities?

- **Location:** Proximity to desirable amenities, such as city centers, metro stations, attractions, and restaurants, significantly influences pricing.
- **Room Type and Capacity (Number of Guests & Bedrooms):** Entire Home/Apt and higher person capacities room generally lead to higher booking amounts.
- **Superhost Status:** Superhost status can positively impact other aspects of a property's performance, such as guest satisfaction and repeat business.
- **Guest Satisfaction (Review Scores):** Properties with high scores receive more positive reviews, which builds trust commands more budget friendly prices.
- **Business Purposes:** Properties booked for business Purposes command higher booking amounts that is the reason why properties with budget friendly are more booked for business purposes.



# AIRBNB ACTIVITY MARKET

## How Do Guest Preferences Differ from One City to Another?

The analysis reveals that guest preferences vary significantly across different European cities. These insights highlight the importance of adapting pricing strategies and marketing efforts based on city-specific guest preferences.

- **Athens, Budapest and Rome:** Guests seek affordable properties (Entire Home/Apt) in historical areas. These cities attract budget travelers, leading to lower average booking prices. Guests prefer Rome as a destination due to Superhost status and multi-listings properties.
- **Amsterdam, London & Paris:** Guests often book private rooms due to the highest booking prices.

## Which areas within the cities experience the highest demand, and what features contribute to their appeal for guests?

- **Popular Areas:** In cities like London, Paris, and Amsterdam, city-center properties are highly sought after due to easy access to major attractions, restaurants, and public transport.
- **Tourist Hotspots:** Areas close to popular tourist landmarks (e.g., Sagrada Familia in Barcelona or the Eiffel Tower in Paris) experience higher demand.
- **Paradisiac Destinations:** Athens, Rome and Budapest are often fully booked due to several factors like satisfaction, lower prices and interesting amenities.



# AIRBNB ACTIVITY MARKET

## What Market Trends and Seasonal Variations Are Observed in Occupancy Rates and Pricing Across Different Cities?

- **Seasonal Variations:** Occupancy rates and pricing typically fluctuate throughout the year, with peak seasons experiencing higher demand and prices.
- **City-Specific Trends:** Market trends and seasonal variations can differ across cities, depending on local factors such as tourism patterns, events, and economic conditions.

## What Insights Can Be Drawn About Future Booking Trends and Potential Revenue Expansion for Hosts?

- Preference for Entire Apartments with affordable booking prices
- Impact of Business Travelling
- Enhance Guest Satisfaction in higher pricing booking cities properties
- The demand for short-term rentals continues to grow, especially in popular tourist destinations.

# RECOMMENDATIONS

## Optimize Pricing Strategies

- Raise the prices incrementally at lower-satisfaction cities, while keeping up with high level quality that drives guest satisfaction
- Spend a portion of the increased revenue on improving guest experience
- Dynamic pricing tools — which raise or lower items based on demand, season and what's going in your neighborhood
- Especially in cities with low booking volume, offer special packages or prices for stays of a longer duration
- Think about increasing the number of listings in areas with budget friendly pricing

## Improve Guest Satisfaction

- Conduct detailed surveys or feedback analysis to pinpoint common issues such as cleanliness, check-in experience, or accuracy of listings. Use this information to develop targeted improvements.
- Introduce a quality control checklist for hosts in cities with lower satisfaction scores
- Provide training and resources to hosts to help them improve their service levels.
- Increase the Visibility of Superhost Listings



**Thank you for your attention**

By Pernel Avougnassou

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